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Geochemical constraints on the origin of secondary magnetizations in the Cambro-Ordovician Royer Dolomite, Arbuckle Mountains, southern Oklahoma

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Abstract

Geochemical and paleomagnetic results from Cambro-Ordovician Royer Dolomite in the Arbuckle Mountains, southern Oklahoma, suggest a connection between dolomite type and magnetizations. One dolomite type, which is most common in the vicinity of the Arbuckle Anticline, has a dull luminescence, low $\delta^{18}O$ values, coeval to slightly lower $^{87}Sr/^{86}Sr$ ratios and contains an easterly Cambro-Ordovician magnetization in magnetite.

To the southeast, closer to the Ouachita Mountain front, most of the Royer dolomite contains a southeasterly and shallow magnetization in magnetite that is interpreted to be Late Paleozoic in age. In contrast to the other dolomites, these dolomites contain abundant authigenic K-feldspar and pyrite, radiogenic $^{87}Sr/^{86}Sr$ ratios, and have a bright (red) luminescence, all of which are consistent with alteration by basinal fluids. At one site, a stromatolitic bioherm, dolomites similar to those at the Arbuckle Anticline are surrounded by the altered dolomites. The distribution of specimen directions at this site is streaked between a Cambro-Ordovician and Late Paleozoic direction.

The results of this study suggest a connection, either direct (by causing precipitation) or indirect (by increasing permeabilities for another fluid), between a radiogenic fluid and a chemical magnetization in magnetite. The relatively unaltered dolomites at the stromatolitic bioherm site escaped the almost pervasive alteration and remagnetization caused by the basinal fluids close to the Ouachita Mountains, a potential source for the fluids.

1. Introduction

Secondary magnetizations that reside in magnetite are common in sedimentary rocks and their origin is controversial (e.g. McCabe and Elmore, 1989; Elmore and McCabe, 1991; Elmore et al., 1993a). Most of these secondary magnetizations are pervasive in rock units and many are synfolding (see McCabe and Elmore, 1989). Several mechanisms have been proposed for these magnetizations, including chemical processes (e.g. McCabe et al., 1989), thermoviscous processes (Kent, 1985), and strain (Hudson et al., 1989).

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Many of these magnetizations are attributed to chemical processes. Evidence commonly cited for this interpretation includes the presence of authigenic magnetite in the rocks (e.g. Suk et al., 1990). Thermal maturities are also too low to permit a thermally activated remagnetization based on blocking temperature–relaxation–time relationships (e.g. Pullaiah et al., 1975; Jackson, 1990). In addition, rock magnetic studies indicate that many of these secondary magnetizations reside in single domain equant grains, without uniaxial anistropy, which is consistent with a diagenetic origin (e.g. Jackson, 1990).

Magnetite authigenesis and acquisition of a chemical remanent magnetization (CRM) has been related to basinal or orogenic fluids (e.g. McCabe et al., 1989), mineralizing fluids (e.g. Symons and Sangster, 1992), hydrocarbons (Elmore and Crawford, 1990), oxidation of siderite (Ellwood et al., 1988), dolomitization (Addison et al., 1985), and maturation of hydrocarbons or other chemical processes not triggered by externally derived fluids (Elmore et al., 1993a).

The action of basinal or orogenic fluids is one of the most popular mechanisms invoked for the apparently pervasive magnetizations (e.g. McCabe and Elmore, 1989). This particularly applies to many Late Paleozoic CRMs, which, along with other phenomena including potassium alteration (e.g. Hearn and Sutter, 1985; Hay et al., 1988), high-temperature alteration of coals (Daniels et al., 1990), and trapping of hot fluids in fluid inclusions (e.g. Leach and Rowan, 1986; Dorobek, 1989) are inferred to be genetically related to the migration of hot brines from the Appalachian and Ouachita fold-thrust belts (e.g. Oliver, 1986, 1992; Bethke and Marshak, 1990). The key evidence for the connection between CRM and orogenic fluids is a spatial association with orogenic belts and a temporal association with orogeny (McCabe and Elmore, 1989). Regional trends in magnetite authigenesis have also been cited as supporting remagnetization by chemically active fluids (e.g. McCabe et al., 1989). Although migration of basinal fluids is a likely agent of remagnetization, the evidence is still largely circumstantial (Elmore and McCabe, 1991). A direct link needs to be made between a CRM and basinal fluids if the secondary magnetizations are to be truly useful as a record of widespread fluid migration.

Geochemical data can provide important information on the origin of fluids that have altered rocks. When used in conjunction with paleomagnetic results, the geochemical data can constrain the origin of magnetizations (e.g. Elmore et al., 1993a). In this study paleomagnetic and geochemical data are used to constrain the origin of secondary magnetizations, and in particular to test for a connection between magnetizations and alteration by basinal fluids, in the Cambro-Ordovician Royer Dolomite in southern Oklahoma.

2. Geologic setting of study area

The Royer Dolomite (lower Arbuckle Group) is a stratal dolomite in the Fort Sill Formation (Ham, 1978). The Royer contains stromatolites, mudcracks, and other sedimentologic features that indicate it was deposited as a shallow marine limestone. Most workers (e.g. Sargent, 1969) identified at least two dolomitization events in the Cambro-Ordovician section in the Arbuckle Mountains. Formation of the stratal units such as the Royer occurred relatively early. A younger event is apparently related to Pennsylvanian-age faults (Sargent, 1969).

The Arbuckle Mountains formed during the Middle to Late Pennsylvanian (Johnson et al., 1988). To the east, deformation of the Ouachita Mountains began in the Middle Pennsylvanian. The buried trace of the Ouachita Mountain front is located just to the east of the Arbuckle Mountains (Fig. 1).

3. Sampling and methods

Paleomagnetic samples were collected from multiple sites at the Dolese Brothers Coleman Quarry in the east, and in a previous study, Nick and Elmore (1990) collected samples on the Arbuckle Anticline in the west (Fig. 1). The magnetization was measured using a Sct cryogenic magnetometer located in a magnetically shielded room. Most specimens were subjected to stepwise

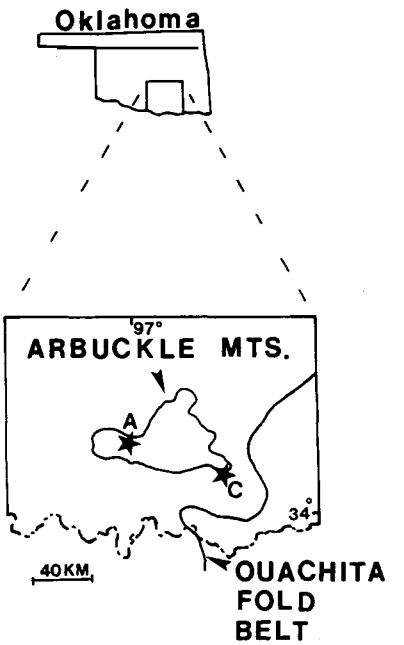


Fig. 1. Sampling localities in the Arbuckle Mountains, southern Oklahoma. Buried trace of Ouachita Mountain front is shown just to the east of Arbuckle Mountain. A, Arbuckle anticline; C, Coleman Quarry.

thermal demagnetization using a Schonstedt TSD-1 oven. Some specimens were subjected to alternating field (AF) demagnetization in a two-axis tumbling unit inside a variable field electromagnet. The demagnetization results were plotted on orthogonal projections, and principal component directions (Kirschvink, 1980) were calculated.

Isothermal remanent magnetization (IRM) acquisition and thermal decay measurements were performed to aid in the identification of the magnetic mineralogy. Specimens were subjected in a stepwise fashion to an increasing IRM up to 0.8 T. The specimens were thermally demagnetized in 13 or more steps.

Petrographic studies (transmitted, cathodoluminescence and reflected light) were conducted on thin sections of chips cut from the paleomagnetic samples. Geochemical data were also obtained from many of the same dolomite samples used in the paleomagnetic investigation as well as additional samples. Details of the procedures used in an isotopic (Sr, C, and O) and trace element investigation are described by Gao et al. (1994).

4. Results and interpretation

4.1. Arbuckle Anticline

In a study of the Royer in the vicinity of the Arbuckle Anticline, Nick and Elmore (1990) report the unit contains two magnetizations: a Cambro-Ordovician magnetization residing in magnetite that occurs in stratal gray dolomite and a Late Paleozoic CRM that resides in hematite and occurs in dedolomitized Royer around karst features. The localized Late Paleozoic CRM is interpreted to have been caused by meteoric fluids.

The gray dolomite has an easterly direction after tilt correction (Fig. 2; Table 1) and clasts of the Royer in the Pennsylvanian Collings Ranch Conglomerate pass a conglomerate test (Fig. 2). This magnetization was interpreted as a detrital remanent magnetization (DRM) or a CRM (Nick and Elmore, 1990).

Petrographic studies of the gray dolomite indicate they are generally coarsely crystalline, usually with subhedral to anhedral fractures. In transmitted light the crystals are clear to slightly cloudy. Under cathodoluminescence, the dolomite is dully to moderately (red) luminescent. Some dolomite crystals, however, have thin, brightly (red) luminescent rims.

Geochemical studies (Gao et al., 1994) indicate that $^{87}Sr/^{86}Sr$ ratios for these dolomites are similar to associated limestones and coeval seawater (Fig. 3), suggesting an early marine origin. Delta ^{18}O values (-7.2 to $-11.9‰$, PDB; Fig. 3) are low, suggesting postdepositional meteoric modification during development of unconformities in the Ordovician (Gao et al., 1994). This is supported by other evidence such as the presence of burial alteration features (e.g. intercrystalline pressure solution and bedding-parallel stylolites) which indicate that the dolomites underwent significant diagenetic alteration (recrystallization) after initial formation (Gao et al., 1994). Similar

alteration histories have been documented for massive stratal dolomites in the Arbuckle Group in the Slick Hills of southwest Oklahoma (Gao, 1990).

4.2. Coleman Quarry

Sampling sites in the Coleman quarry were collected over a stratigraphic range of 60 m. Although the rocks are all dolomitized, several different facies are present. Sites were collected in dolomitized mudstone, coaserly crystalline dolomite, quartz-rich dolomitized mudstone, and a dolomitized stromatolitic bed.

Paleomagnetic results from the Coleman Quarry indicate a streaked easterly to southeasterly distribution of specimen directions. Orthogonal projections of the remanent magnetization vectors for most specimens from the Coleman Quarry show linear removal of a southeasterly and shallow direction (Fig. 4A), in some cases after removal of a northerly, steep down component interpreted to be a modern magnetization of viscous origin. Specimens from eight of nine sites

Table 1
Paleomagnetic results from the Royer Dolomite, southern Oklahoma

Site no.	N/no.	D (in situ)	I (in situ)	k	α_{95}	D (tilt-adjusted)	I (tilt-adjusted)	
Coleman Quarry								
RD33	5/5	147.4	−0.1	90	8.1	144.9	21.0	
RD34	10/10	139.8	3.0	182	3.6	135.9	22.1	
RD35	7/7	150.0	23.0	33	10.6	139.1	44.6	
RD40	9/13	141.5	−2.4	92	5.4	139.5	17.3	
RD32	9/10	145.8	1.9	196	3.7	142.6	22.6	
RD31	5/6	146.1	−1.5	551	3.3	143.4	20.7	
RD37A	3/3	137.1	8.9	160	9.8	131.3	26.9	
RD37	4/4	146.1	7.0	39	15.0	141.5	27.6	
RD1E	17/20	128.3	10.9	22	7.8	118.0	28.4	
						Paleomagnetic pole		
						°E	°N	dp/dm
Location means								
Coleman (in situ)	8/8	144.2	5.0	76	6.4	133	40 [a]	3.2/6.4
Coleman (tilt-corrected)	8/8	139.8	25.4	75	6.4	129	30 [a]	3.7/6.9
Arbuckle Anticline (in situ, specimen directions)	32/36	110.6	14.1	27	4.8	154.9	12.6 [a]	2.5/4.9
Arbuckle Anticline (tilt-corrected, specimen directions)	32/36	109.2	10.5	33.3	4.3	157.3	12.6 [a]	2.2/4.4

D and I are the mean declination and inclination. N is the number of specimens used compared with the number of specimens demagnetized (no.). k is the Fisher (1953) precision parameter and α_{95} is the cone of 95% confidence about the mean direction. dp and dm are the radii of the oval of confidence about the paleomagnetic pole in the colatitude direction and at right angles to it, respectively.

[a] Denotes that the pole is the antipodal equivalent of the calculated pole.

were found to be dominated by the southeasterly and shallow component (Fig. 5). The mean (before tilt correction) of the site means is 144.2° declination and 5.0° inclination ($k = 76$, $\alpha_{95} = 6.4°$; Table 1).

The specimen directions from one site at the Coleman Quarry, a dolomitized stromatolitic bioherm (site 1E), are different to those from the other sites (Table 1). After removal of a northerly and steep down component at low temperatures or field strengths, easterly to southeasterly directions were removed from 17 specimens (Fig. 4(B)). The specimen directions from this site account for the overall streaked distribution of directions from the Coleman Quarry (Fig. 5). Decay of the magnetizations in a few specimens revealed a hint of multicomponent behavior with removal of a southeasterly and shallow direction at low field strengths and the apparent removal of an easterly and shallow direction at higher strengths (Fig. 4(C)).

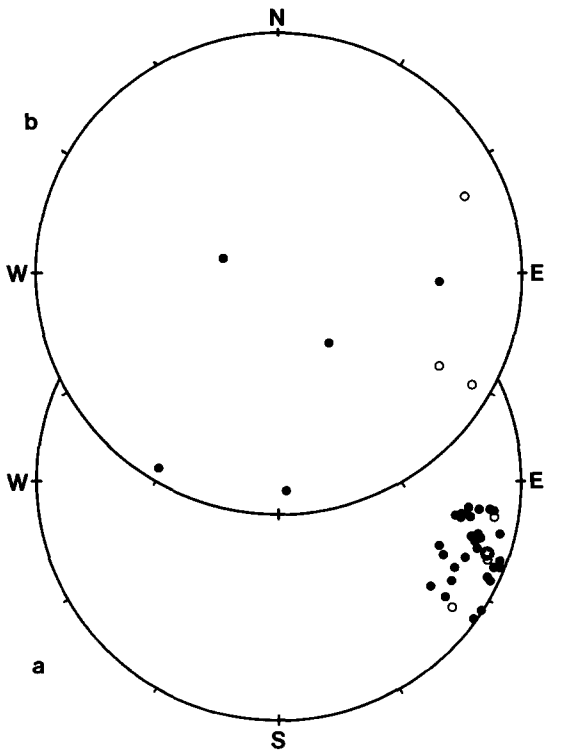


Fig. 2. Paleomagnetic results (specimen directions) from gray dolomite collected in the Arbuckle Anticline vicinity (a) and mean directions for clasts (centers) of the Royer in the Pennsylvanian Collings Ranch Conglomerate (b). From Nick and Elmore (1990).

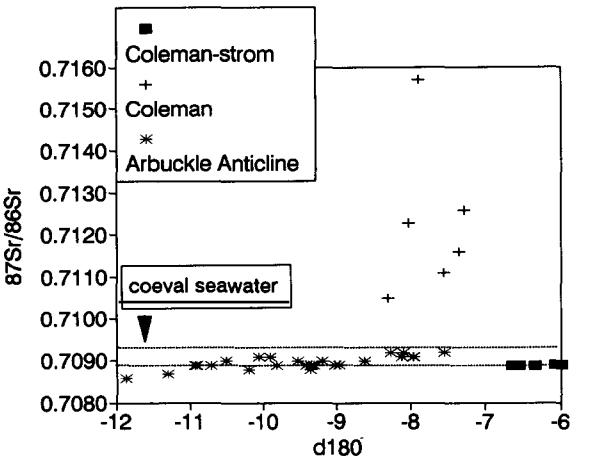


Fig. 3. $^{87}Sr/^{86}Sr$ vs. $\delta^{18}O$ plot, highlighting that (1) gray dolomites from the Arbuckle Anticline (stars) with the Cambro-Ordovician magnetization have low $\delta^{18}O$ values and coeval Late Cambrian-Early Ordovician $^{87}Sr/^{86}Sr$ ratios (≈ 0.7089–0.70092), (2) dolomites from the Coleman Quarry (pluses) that contain a Late Paleozoic magnetization have elevated (radiogenic) $^{87}Sr/^{86}Sr$ ratios (Burke et al., 1982), and (3) the stromatolitic bed of the Coleman Quarry with the Cambro-Ordovician magnetization has $^{87}Sr/^{86}Sr$ ratios (squares) similar to the gray dolomites at the Arbuckle Anticline.

This site (1E) has a distinct magnetization compared with the other sites at the Coleman Quarry. Compared with specimens from other sites, the intensity of magnetization in specimens from site 1E is lower by approximately an order of magnitude. Most specimen directions are also more easterly both before (Fig. 5) and after tilt correction than for specimens from the other Coleman sites. The directions are similar to those from the Royer at the Arbuckle Anticline although with a higher positive inclination after tilt-correction (Fig. 5). Despite the streaked distribution of specimen directions, means were calculated for this site (Table 1).

The IRM acquisition of representative specimens from all sites at the Coleman Quarry shows near complete magnetic saturation at 0.2 T (Fig. 6). The IRM saturation intensity for specimens from site 1E is lower compared with specimens

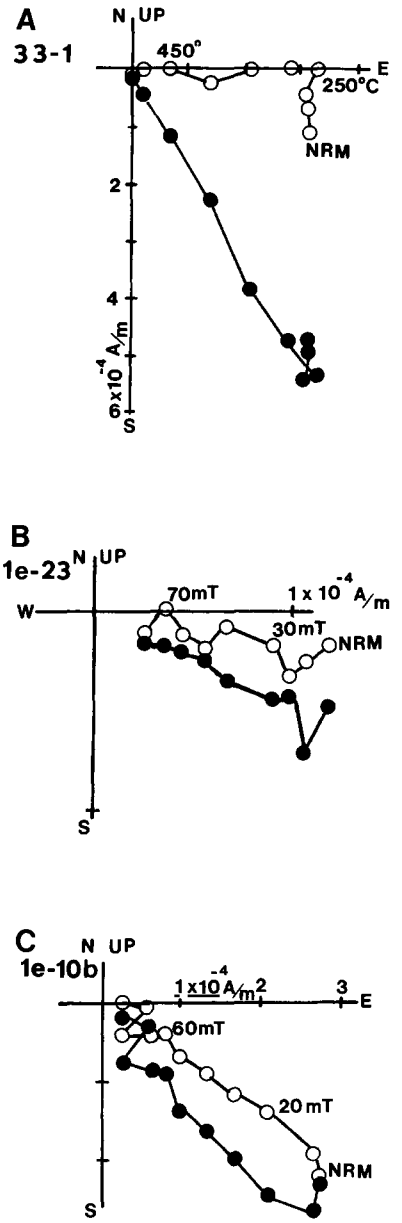


Fig. 4. Orthogonal projections of the demagnetization vector for representative samples from the dolomite at the Colman Quarry. (A) Specimen from the altered dolomites. Note removal of southeasterly and shallow direction. (B) Specimen from the stromatolitic bed. Note removal of east–southeasterly direction. (C) Specimen from the stromatolite bed. Note removal of a southeasterly direction at moderate field strengths (20–60 mT) and an easterly direction at high field strengths (60–80 mT). Open symbols, vertical component; closed symbols, horizontal component.

from other sites. Thermal demagnetization removed the IRM at 600°C in most specimens (Fig. 7). Comparison of site 1E specimens with specimens from other sites indicates similar thermal demagnetization curves, although the specimens from site 1E have a more distributed range of unblocking temperatures (Fig. 7). AF decay of representative specimens to 125 mT removed approximately 90% of the IRM and there were no obvious differences in the curves between site 1E specimens and the other specimens. The demagnetization characteristics, supported by the rock magnetic studies, indicate that the magnetizations in the specimens are dominated by a low coercivity magnetic mineral that decays below 600°C. This suggests magnetite is the dominant magnetic carrier in these specimens.

Samples which contain the southeasterly component are brightly (red) luminescent and contain abundant authigenic K-feldspar and pyrite. The crystals are also cloudy. The samples also have high Fe and Mn concentrations (Fig. 8), high (radiogenic) $^{87}Sr/^{86}Sr$ ratios (0.07105–0.7157; Fig. 3), and $\delta^{18}O$ values (Fig. 3) in the range of -7.3 to $-8.3‰$ (PDB). Basinal fluids are commonly enriched in ^{87}Sr as well as Fe and Mn as a result of silicate water contamination (Stueber et al., 1984; and many others). The geochemical and petrographic characteristics of these dolomites suggest alteration by basinal fluids.

Petrographic and geochemical characteristics of site 1E with the more easterly specimen directions are different to those at the other sites at the Coleman Quarry, and are similar to the Royer at the Arbuckle Anticline. $^{87}Sr/^{86}Sr$ ratios are coeval for Cambro-Ordovician seawater and not radiogenic (Fig. 3). Fe and Mn concentrations are low (Fig. 8) compared with the altered dolomites at Coleman. Oxygen isotope values are also low (-6.0 to $-6.6‰$, PDB), although not as low as for the Royer at the Arbuckle Anticline (Fig. 3). The rocks from this site also have a dull (red) luminescence.

Interpreting the ages of the southeasterly and easterly components of magnetization at the Coleman Quarry is difficult owing to the lack of field tests. The streaked distribution of specimen directions from site 1E suggests either remanence

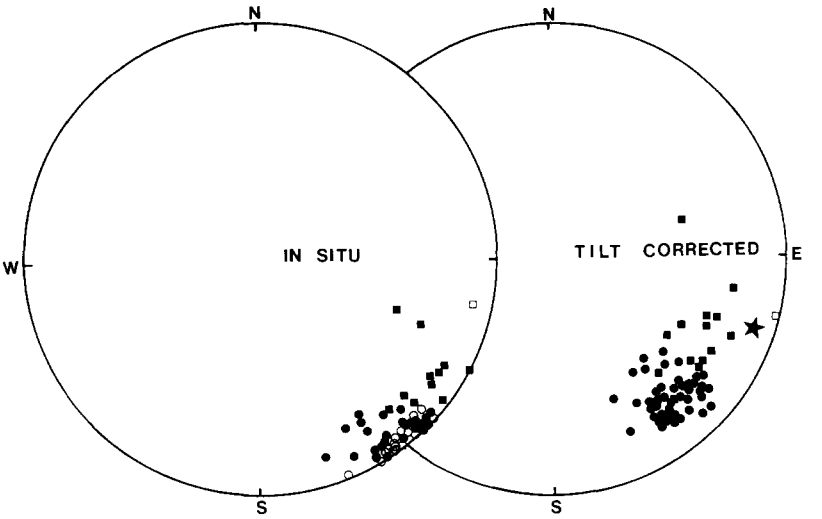


Fig. 5. Equal area stereonets with paleomagnetic results from the Coleman Quarry. Circles show specimens with the southeasterly and shallow direction. Squares are specimen directions from the stromatolitic bed. The mean for the gray dolomite (star) at the Arbuckle Anticline is also shown. Closed symbols, lower hemisphere; open symbols, upper hemisphere.

acquisition over an interval of time with significant apparent polar wander, or, the vector addition of two (or more) components. Because the magnetization in some specimens is apparently composed of both the southeasterly and easterly components (Fig. 4(C)), a vector addition is the most likely case. In specimens with the two components, the coercivities (or blocking temperatures) of the components do not completely overlap, which is in contrast to most of the specimens

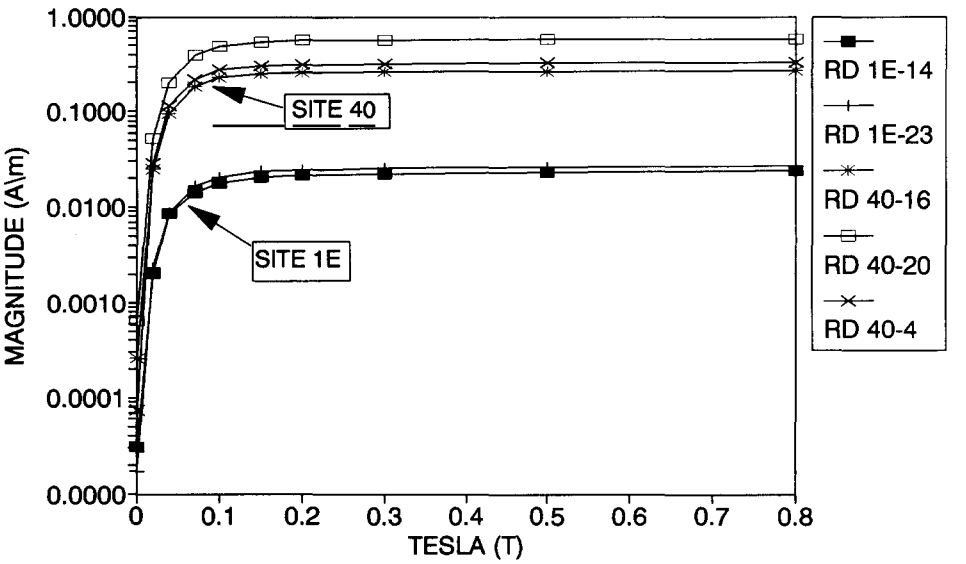


Fig. 6. Isothermal remanent magnetization (IRM) acquisition curves for representative specimens of both dolomite types at the Coleman Quarry. Site 40 specimens are representative of specimens from all sites except 1E.

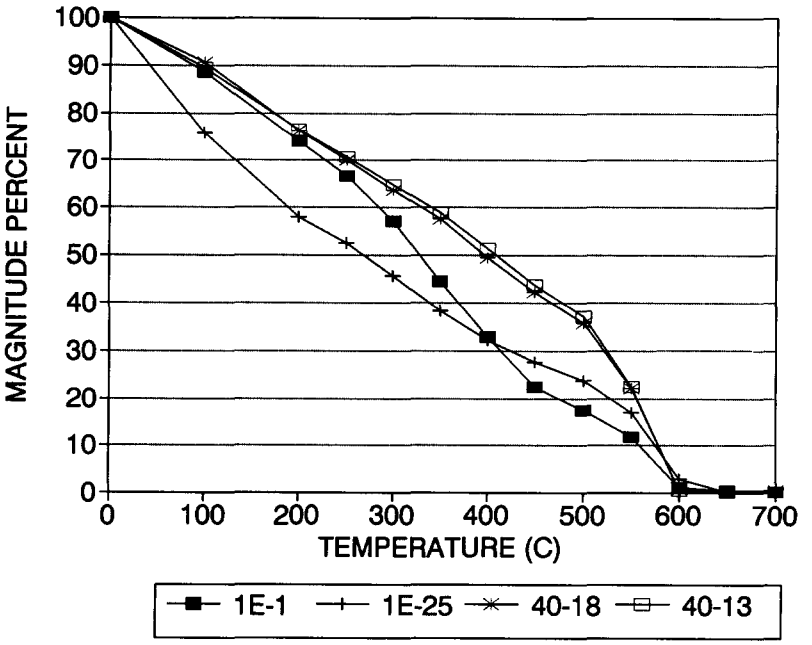


Fig. 7. Thermal demagnetization of the IRM for representative specimens. Note that specimens from site 1E have a more distributed range of unblocking temperatures compared with specimens from site 40 which undergo a higher percentage of decay at higher temperatures. IRM saturation intensities in A m^{-1} for each specimen are 1E-1 = 0.045, 1E-25 = 0.073, 40-13 = 0.555, 40-18 = 0.355.

from site 1E. The similarity in directions between the results from the Arbuckle Anticline and site 1E suggests that the Cambro-Ordovician direc-

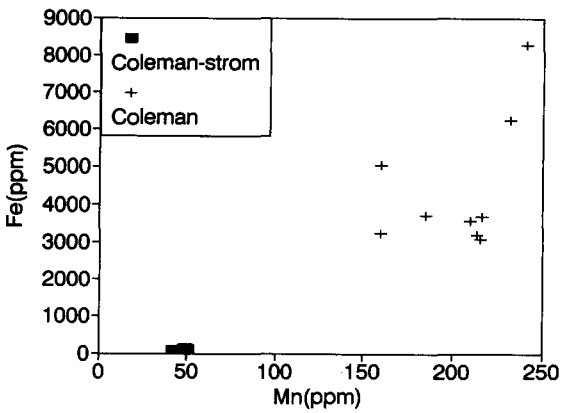


Fig. 8. Fe vs. Mn plot showing that most sites from the Coleman Quarry with the Late Paleozoic CRM (pluses) have high amounts of Fe and Mn whereas the stromatolitic dolomite site with Cambro-Ordovician magnetization (squares) has lower amounts of Fe and Mn.

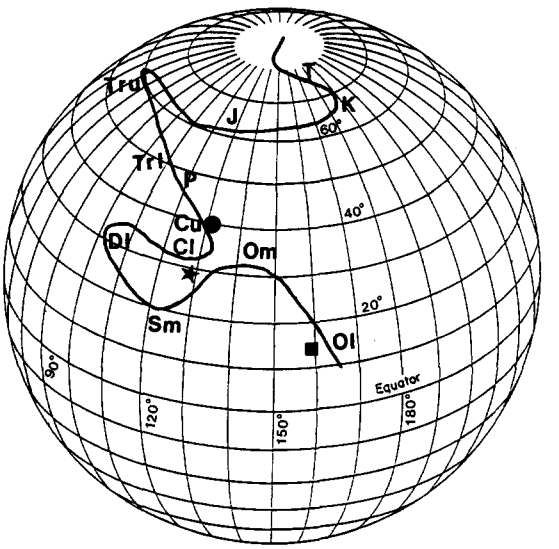


Fig. 9. North American apparent polar wander path (Van der Voo, 1989) with the poles for the gray dolomite at the Arbuckle Anticline (square, after tilt-correction) and from altered dolomite at Coleman Quarry (circle, before tilt-correction; star, after tilt-correction).

tion may be one endmember of a vector addition. The similarity in geochemical results from the two areas is consistent with this interpretation. The other endmember is the southeasterly direction present in the majority of the specimens at Coleman.

The before tilt-corrected paleopole for the southeasterly component (133°E longitude and 40°N latitude) plots on the upper Pennsylvanian-lower Permian portion (Fig. 9) of the North American apparent polar wander path (APWP) whereas the after tilt-corrected pole (129°E, 29°N) plots between the Pennsylvanian and Silurian segments of the path. The lack of a field test for this component makes an independent assessment of the age of remanence acquisition difficult. Based on the pole positions only, the magnetization could be Pennsylvanian–Permian or perhaps Silurian in age.

5. Discussion and conclusions

The magnetizations in the Royer are not thermoviscous in origin because burial temperatures were too low. Maximum burial temperature in the Royer were less than 100°C based on a maximal burial depth of 2.16 km (Ham, 1955) and a geothermal gradient of 30°C km^{-1}. A burial temperature of < 100°C is consistent with burial temperature estimates based on vitrinite reflectance data from the Arbuckle Mountains (Cardott et al., 1990).

The paleomagnetic and petrographic/geochemical results suggest a relation between the magnetizations and dolomite type. Dolomite that has been altered by basinal fluids contains a CRM that is distinct from the magnetization in relatively less altered dolomite. The basinal fluids were radiogenic, and added Fe and Mn to the dolomite. The basinal fluids could have directly caused precipitation of magnetite although there is no evidence for such a direct connection. Alternatively, the alteration may have provided conduits for other fluids such as hydrocarbons which could have caused precipitation of authigenic magnetite (e.g. Elmore and Crawford, 1990).

Although the results from the Coleman Quarry are consistent with a connection between the CRM with a southeasterly direction and alteration by basinal fluids, there is uncertainity regarding the time of remanence acquisition for the CRM. Based on the pole positions, the CRM could be Late Pennsylvanian–Permian or Silurian in age. The geochemical results, however, may provide some constraints as to the age of the CRM. High (radiogenic) $^{87}Sr/^{86}Sr$ ratios appear to have characterized fluids derived from the Arkoma Basin and/or the Ardmore Basin adjacent to the Arbuckle Mountains (Gao et al., 1994). The geometries necessary for migration, and the faults and fractures which provided conduits for flow in the Arbuckle Mountains were not created until the Late Paleozoic (Johnson et al., 1988). This evidence suggests that migration of basinal fluids and the associated dolomite alteration did not occur until the Late Paleozoic. Although the argument would obviously be stronger if there were a paleomagnetic test to independently assess the age of the magnetization, the connection between dolomite alteration and the CRM suggests that remanence acquisition occurred in the Late Pennsylvanian–Permian. This interpretation is also consistent with a study by Elmore et al. (1993b) that presents geochemical and paleomagnetic results from Ordovician carbonates in the Arbuckle Mountains indicating that basinal fluids caused acquisition of Permian CRM in hematite around the conduits for flow.

Dolomite from the Arbuckle Anticline that has a Cambro-Ordovician geochemical signature and was not altered by the basinal fluids contains a Cambro-Ordovocian paleomagnetic direction. The fact that one site at the Coleman Quarry which has a Cambro-Ordovician geochemical signature also contains directions that are similar to the results from the Arbuckle Anticline supports the connection between dolomite type and magnetization. The dolomite at this site escaped, perhaps owing to original differences in permeability, the almost pervasive remagnetization at the Coleman Quarry. The mechanism of remanence acquisition for the Cambro-Ordovician component is open to question, but there is no evidence it involved basinal fluids. Nick and El-

more (1990) reported that the Royer contains spheres interpreted to be authigenic magnetite, but the spheres are probably too coarse to carry the remanence (e.g. Jackson, 1990). The magnetization could be a DRM or a CRM.

The results of this study suggest a link between a CRM in magnetite and alteration by basinal fluids. This is important for the origin of secondary magnetizations in magnetite, particularly in light of other recently published investigations which question the role of basinal fluids as agents of remagnetization for CRM in magnetite. For example, in a study on the Viola Limestone by Elmore et al. (1993a), the unit was found to contain a Permian CRM in hematite around mineralized fractures and a pervasive Pennsylvanian CRM in magnetite away from the fractures. There is no geochemical evidence that externally derived fluids entered the unit except for around the fractures which were the conduits for flow. Another geochemical mechanism, therefore, not triggered by externally derived fluids and perhaps mediated by relatively low burial temperatures, may be responsible for the pervasive remagnetization. Possible mechanisms include the maturation of hydrocarbons which occurs between 65°C and $\approx$ 150°C (Tissot and Welte, 1978) or the smectite-to-illite transformation which occurs below 100°C (e.g. Boles and Franks, 1979).

The results from the Royer Dolomite suggest a connection between basinal fluids, perhaps triggered as a result of tectonic activity in the Late Paleozoic, and a CRM in magnetite. A determination of the exact mechanisms of remagnetization, however, remains unresolved and should be the focus of future research. Geochemical and paleomagnetic studies testing for connections between remagnetization and specific fluids around conduits for flow could be productive in this regard.

Acknowledgments

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